

HTML - Colors

HTML Colors

- Colors in HTML can be specified using:
 1. **Color Names** (e.g., red, blue, green).
 2. **Hexadecimal Values** (e.g., #FF0000 for red).
 3. **RGB Values** (e.g., rgb(255, 0, 0) for red).
 4. **RGBA Values** (e.g., rgba(255, 0, 0, 0.5) for semi-transparent red).
 5. **HSL Values** (e.g., hsl(0, 100%, 50%) for red).
 6. **HSLA Values** (e.g., hsla(0, 100%, 50%, 0.5) for semi-transparent red).

1. Color Names

- HTML supports 140 standard color names.
- Example:

```
<p style="color: red;">This text is red.</p>
<p style="background-color: blue;">This background is blue.</p>
```

2. Hexadecimal Values

- Colors are represented as a combination of **red, green, and blue (RGB)** in hexadecimal format.
- Format: #RRGGBB, where RR, GG, and BB are hexadecimal values (00 to FF).
- Example:

```
<p style="color: #FF0000;">This text is red.</p>
<p style="background-color: #0000FF;">This background is blue.</p>
```

3. RGB Values

- Colors are defined using the rgb() function, which takes three values for red, green, and blue (0 to 255).
- Example:

```
<p style="color: rgb(255, 0, 0);">This text is red.</p>
<p style="background-color: rgb(0, 0, 255);">This background is blue.</p>
```

4. RGBA Values

- Similar to RGB but includes an **alpha channel** for transparency (0 = fully transparent, 1 = fully opaque).
- Example:

```
<p style="color: rgba(255, 0, 0, 0.5);">This text is semi-transparent red.</p>
<p style="background-color: rgba(0, 0, 255, 0.3);">This background is semi-transparent blue.</p>
```

5. HSL Values

- Colors are defined using **hue, saturation, and lightness**:

- **Hue:** A degree on the color wheel (0 = red, 120 = green, 240 = blue).
- **Saturation:** Percentage (0% = gray, 100% = full color).
- **Lightness:** Percentage (0% = black, 100% = white).
- Example:

```
<p style="color: hsl(0, 100%, 50%);">This text is red.</p>  
<p style="background-color: hsl(240, 100%, 50%);">This background is blue.</p>
```

6. HSLA Values

- Similar to HSL but includes an **alpha channel** for transparency (0 = fully transparent, 1 = fully opaque).
- Example:

```
<p style="color: hsla(0, 100%, 50%, 0.5);">This text is semi-transparent red.</p>  
<p style="background-color: hsla(240, 100%, 50%, 0.3);">This background is semi-transparent blue.</p>
```

Common Uses of Colors in HTML

1. Text Color:

```
<p style="color: green;">This text is green.</p>
```

2. Background Color:

```
<p style="background-color: yellow;">This background is yellow.</p>
```

3. Border Color:

```
<p style="border: 2px solid purple;">This has a purple border.</p>
```